

Comparative Evaluation of RGB-ToF Fusion Methods for Improving SLAM

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Abstract—This paper presents a comparative evaluation of RGB-Time-of-Flight (ToF) fusion methods for depth reconstruction and downstream SLAM. We study a broad set of approaches, including nearest-neighbor upsampling, RGB-guided weighted median filtering, confidence-aware completion based on ToF amplitude, multi-frame fusion, plane-based refinement, superpixel-guided planar completion, variational weighted least squares, and neural monocular-depth priors. The methods are evaluated on synchronized RGB-ToF datasets with calibrated sensor alignment, and the main benchmark is performed on `dod_1500us_20MHz`, where dense ground-truth depth is available. Depth quality is measured using overlap-only, full-frame, and filled-region benchmarks, while downstream geometric usefulness is evaluated in an Open3D RGB-D SLAM pipeline and in a mapping setting with known camera trajectory. The results show that the best pixel-level depth method is not automatically the best SLAM input. Variational weighted least squares achieved the strongest reproduced overlap-only depth benchmark, whereas more geometry-conservative plane-based methods produced the most accurate SLAM trajectories. Neural monocular-depth priors generated visually plausible dense maps but were less reliable for metric RGB-D SLAM in the evaluated setup. Overall, the study shows that RGB-ToF fusion should be assessed not only as a depth restoration problem, but also as a full robotics perception pipeline component.

Index Terms—RGB-D SLAM, Time-of-Flight, depth completion, depth fusion, mapping, Open3D

I. INTRODUCTION

Reliable depth perception is a fundamental requirement for many robotics applications, including scene understanding, obstacle avoidance, navigation, and Simultaneous Localization and Mapping (SLAM). In practical systems, depth information is often obtained from active sensors such as Time-of-Flight (ToF) cameras, while RGB cameras provide high-resolution appearance information with sharp object boundaries and rich texture. Since these sensing modalities have complementary properties, their joint use is a natural direction for improving both depth reconstruction and downstream geometric perception.

Despite their practical value, ToF sensors suffer from several well-known limitations. Their measurements are typically lower in spatial resolution than RGB images and are often degraded by noise, missing values, multipath interference, boundary distortions, and reduced reliability on reflective or weakly illuminated surfaces. As a result, raw ToF depth maps are often too sparse or unstable to be directly used in high-quality reconstruction and SLAM pipelines. In contrast, RGB images preserve detailed spatial structure, but do not provide metric depth on their own. This motivates the use of RGB-

guided depth enhancement and RGB-ToF fusion methods that aim to denoise, refine, and complete raw ToF depth maps.

A broad range of fusion strategies can be used for this purpose. Local RGB-guided filters can suppress outliers while preserving edges, confidence-aware completion can exploit ToF signal quality information to reconstruct missing values, temporal fusion can aggregate information across neighboring frames, and superpixel-based geometric models can enforce piecewise-planar structure in the recovered depth. However, improved pixel-wise depth quality does not automatically imply improved SLAM performance. A depth map may achieve better reconstruction metrics while still introducing artifacts that reduce geometric consistency, weaken registration, or distort the final map. Therefore, evaluating RGB-ToF fusion only at the level of depth reconstruction is insufficient for robotics applications.

The goal of this project is to compare a broad family of RGB-ToF fusion methods, quantify their depth-reconstruction quality, and then test whether these improvements carry over to downstream SLAM and mapping. In addition to classical fusion modules, we also consider variational depth regularization and neural monocular-depth priors. The main contributions are:

- a unified benchmark for classical, variational, and neural RGB-ToF fusion methods;
- a depth-evaluation protocol that separates overlap-region accuracy from hole-filling quality;
- a project-specific Open3D RGB-D SLAM evaluation based on trajectory error relative to the strongest available depth reference;
- a known-trajectory mapping study that isolates mapping quality from odometry drift.

II. RELATED WORK

Depth enhancement for active range sensors is a well-established problem in computer vision and robotics. Low-resolution and noisy depth measurements are commonly refined using information from a registered RGB image, which provides high-frequency spatial structure and more reliable object boundaries. A widely used family of methods is based on edge-aware local filtering, where depth values are smoothed or propagated according to spatial distance and color similarity. Such methods preserve discontinuities better than conventional linear smoothing and are particularly effective for removing local outliers while avoiding excessive boundary bleeding.

Another important direction is confidence-aware depth completion. In active sensors, not all depth measurements are equally reliable, and auxiliary quantities such as amplitude or confidence maps can be used to distinguish stronger observations from uncertain ones. Confidence-weighted completion methods use this information to guide interpolation and reconstruction, allowing the algorithm to trust stable measurements more strongly while filling invalid regions conservatively. These methods are especially relevant for ToF cameras, where depth quality depends strongly on signal strength and surface properties.

Temporal fusion methods extend depth refinement from a single frame to a short image sequence. By aggregating measurements from neighboring frames, they can suppress random noise, reduce temporal flicker, and increase valid depth coverage. In the simplest form, this aggregation is performed directly in image space using temporal and appearance-based weights. More advanced formulations incorporate explicit geometric reprojection between frames, but such methods require accurate pose estimation and increase computational complexity. In practical pipelines, temporal fusion is therefore often used as a lightweight denoising and completion stage.

Geometric priors have also been used for depth completion. In many indoor scenes, large regions can be approximated as piecewise-planar surfaces. This motivates methods based on image segmentation, superpixels, or planar model fitting, where missing depth values are reconstructed from a locally fitted surface rather than from purely local averaging. Such approaches can recover structured regions more accurately than generic smoothing methods, although their performance depends on segmentation quality and on the validity of the planar assumption.

The effect of depth quality on SLAM has also been studied from both geometric and visual perspectives. Depth-only pipelines such as ICP-based SLAM rely heavily on the geometric consistency of point clouds and are sensitive to outliers, misalignment, and oversmoothed surfaces. Visual odometry methods, in contrast, exploit image features and are less dependent on dense depth quality, but still benefit from accurate depth for 3D reconstruction and scale-consistent mapping. Hybrid pipelines that combine feature-based visual motion estimation with depth-based geometric refinement can often provide better robustness than purely visual or purely geometric methods.

These observations motivate the present study. Rather than evaluating RGB-ToF fusion only as an image restoration problem, we analyze it as a preprocessing stage for SLAM and compare how different reconstruction strategies affect both depth benchmarks and downstream mapping performance.

III. METHODOLOGY

A. Datasets and Preprocessing

The main benchmark dataset is `dod_1500us_20MHz`. Additional experiments were also run on `wooden_1500us_20MHz`, `dod_750us_20MHz`, and on a merged-exposure `750us+3500us` set. Each sequence

provides rectified RGB images, rectified ToF depth maps, ToF amplitude images, and a calibrated RGB-ToF setup. For `dod_1500us_20MHz`, dense ground-truth depth is available and is used for quantitative benchmarking.

The preparation pipeline is shown in Fig. 1. Calibration parameters are first estimated and then used for undistortion and rectification of both RGB and ToF streams. After rectification, frames are paired by file stem to form the final aligned dataset used in all further experiments.

Frames are matched by file stem. Raw ToF is considered valid only when depth is positive and the corresponding amplitude exceeds a threshold. Depending on the downstream method, RGB and depth are either aligned on the RGB grid (upsampled depth) or on the ToF grid (RGB resized to ToF resolution). This distinction is important because our study evaluates not only fused depth quality but also how the chosen sampling grid affects SLAM and mapping.

B. Fusion Method Families

1) *Baseline upsampled ToF*: The baseline is nearest-neighbor upsampling of rectified ToF depth to RGB resolution. It introduces no new geometry and is therefore a natural reference for overlap-only benchmarking and for testing whether more complex fusion actually helps SLAM.

2) *RGB-guided Weighted Median Filtering*: Weighted median filtering (WMF) denoises the raw ToF map while preserving edges guided by RGB. For a local neighborhood $\mathcal{N}(p)$ of pixel p , each candidate depth value d_q is assigned a weight

$$w_{pq} = \exp\left(-\frac{\|p - q\|^2}{2\sigma_s^2}\right) \exp\left(-\frac{\|I_p - I_q\|^2}{2\sigma_c^2}\right), \quad (1)$$

where I_p is the RGB color at pixel p . The fused value is the weighted median of the neighborhood depths rather than a weighted average, which improves robustness to isolated ToF outliers.

3) *Confidence-aware Completion*: ToF amplitude is used as a confidence proxy. Valid high-amplitude pixels are trusted more strongly than weak or missing measurements. Missing values are filled iteratively by weighted aggregation of nearby confident pixels:

$$\hat{d}_p = \frac{\sum_{q \in \mathcal{N}(p)} w_{pq} c_q d_q}{\sum_{q \in \mathcal{N}(p)} w_{pq} c_q}, \quad (2)$$

where c_q is normalized amplitude-based confidence. Newly created values receive reduced confidence in later iterations so that synthetic depth does not dominate the estimate.

4) *Plane-based Completion*: Two families of planar regularization were used. The first applies confidence-aware completion followed by local plane refinement; the second fits planes inside RGB superpixels. In both cases, a local depth model

$$d(x, y) = ax + by + c \quad (3)$$

is estimated from valid measurements. The plane is accepted only if residual and slope constraints are satisfied. This introduces a structural prior that is especially useful for flat surfaces and polyhedral objects.

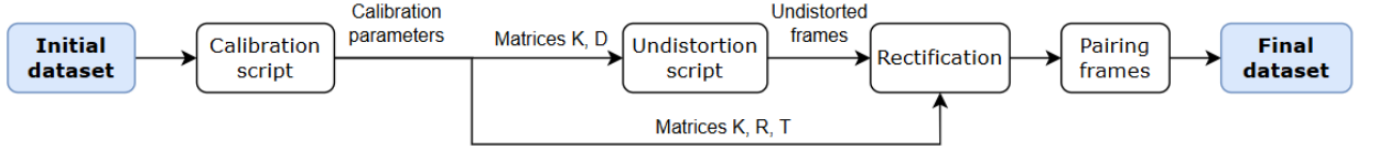


Fig. 1. Dataset preparation pipeline. Calibration parameters are first estimated, then used for undistortion and rectification, after which RGB and ToF frames are paired into the final aligned dataset.

5) *Multi-frame Fusion*: Multi-frame fusion uses a temporal window around the current frame. Neighboring observations are weighted by temporal distance, RGB similarity, and confidence:

$$w_{pqt} = \exp\left(-\frac{\Delta t^2}{2\sigma_t^2}\right) \exp\left(-\frac{\|I_p^t - I_p^{t_0}\|^2}{2\sigma_c^2}\right) c_p^t. \quad (4)$$

In the current implementation, fusion is performed directly in image space, without explicit motion-compensated reprojection, so it acts as temporal denoising and soft completion rather than strict multi-view reconstruction.

6) *Combined Pipelines*: We evaluated several combinations of the above blocks: confidence completion followed by WMF, confidence completion followed by multi-frame fusion, and a three-stage confidence + multi-frame + superpixel pipeline. These pipelines were designed to first increase coverage, then stabilize the result, and finally impose geometric structure.

7) *Variational WLS*: Variational weighted least squares (Var-WLS) solves for a refined depth field d by minimizing a confidence-weighted data term and an RGB-guided smoothness term:

$$E(d) = \sum_i c_i (d_i - z_i)^2 + \lambda \sum_{(i,j) \in \mathcal{E}} g_{ij} (d_i - d_j)^2, \quad (5)$$

where z_i is the raw ToF depth, c_i is confidence, and g_{ij} decreases across strong RGB edges. This model favors smooth depth inside homogeneous regions while preserving discontinuities aligned with color structure.

8) *Neural Depth Priors*: We also tested neural monocular-depth estimators: AnyDepth, DepthAnything, and SelfToF. These methods do not rely on local RGB-ToF filtering but instead predict dense depth from learned image priors. To compare them fairly against ground-truth depth, their outputs were benchmarked either directly (if already on the RGB grid) or after scaling / resizing to the benchmark resolution.

C. Current SLAM Pipeline

The main downstream SLAM algorithm in the project is the Open3D-based pipeline implemented in `hybrid_rgbd_slam.py`. Despite the filename, this is not a feature-based SLAM system such as ORB-SLAM. Instead, it is a classical RGB-D odometry and mapping pipeline:

- 1) build aligned RGB-D frames;
- 2) estimate frame-to-frame camera motion;
- 3) accumulate a global camera trajectory;

4) back-project each depth frame into 3D using the estimated pose;

5) merge all point clouds into a global map.

Open3D estimates motion directly from RGB-D alignment rather than from explicit keypoint descriptors. Conceptually, it searches for the rigid transform $T_{k,k+1}$ that best aligns consecutive frames by minimizing a mixture of photometric and geometric mismatch:

$$T_{k,k+1} = \arg \min_T \sum_i \|I_{k+1}(\pi(TX_i)) - I_k(\pi(X_i))\|^2 + \alpha r_d(T)^2, \quad (6)$$

where X_i are 3D points obtained from the current depth map, $\pi(\cdot)$ is projection to the image plane, and r_d is a depth-consistency residual. The estimated transformations are chained to obtain the full camera trajectory. The final output is a colored point cloud map and a trajectory file.

D. Mapping with Known Trajectory

To separate mapping quality from odometry quality, we also implemented a dedicated mapping mode with known camera poses in `known_trajectory_mapping.py`. In this mode, the camera follows a prescribed linear path from

$$(0, 0, 0) \rightarrow (0.06252, -1.02404, -0.002)$$

with a fixed global orientation. No odometry is estimated from the data. Each depth frame is simply back-projected to 3D and transformed by the corresponding known pose. This produces a direct mapping test: if the depth is geometrically consistent, the polyhedron should appear sharp; if the depth is noisy or biased, the shape becomes blurred or elongated.

E. Evaluation Protocol

Depth quality is benchmarked against ground-truth depth using coverage, MAE, RMSE, AbsRel, and threshold accuracy. Three evaluation modes are used:

- *pred-valid* (full prediction quality),
- *overlap-only* (quality only where both prediction and baseline are valid),
- *filled-region-only* (quality of newly reconstructed pixels).

For SLAM, the best available reference trajectory is the SLAM run with ground-truth depth (`gt_depth`). Since the same odometry code is used but with the strongest possible depth input, we treat this trajectory as the project-specific upper bound. Other SLAM trajectories are evaluated against it using mean position error, RMSE, and final-frame error. This makes the comparison practical and consistent with the rest of the project.

IV. EXPERIMENTAL SETUP

A. Datasets

The main quantitative study uses `dod_1500us_20MHz`. This dataset contains synchronized RGB, rectified ToF, amplitude maps, calibration, and dense ground-truth depth. We additionally validated parts of the pipeline on `wooden_1500us_20MHz`, `dod_750us_20MHz`, and a merged-exposure `750us+3500us` dataset to test transfer to different scenes and signal conditions.

B. Evaluated Depth Representations

The benchmark includes raw upsampled ToF, WMF variants, confidence-aware completion, confidence+plane refinement, confidence+WMF, multi-frame fusion, superpixel-plane fitting, confidence+multi-frame+superpixel pipelines, variational WLS, and neural baselines (AnyDepth, DepthAnything, SelfToF). For ToF-resolution methods, both native and upsampled outputs were kept. RGB-resolution outputs were used for direct comparison to ground-truth depth and for the RGB-grid SLAM experiments.

C. Depth Benchmark Metrics

Classical fusion methods were compared primarily on `overlap-only` metrics, because this mode measures how well a method refines the part of the scene where the original ToF already provides support. This avoids mixing two effects: denoising of reliable pixels and aggressive hole filling. Filled-region performance was also examined separately for methods whose main purpose is to extend coverage.

D. SLAM and Mapping Metrics

For SLAM, we report trajectory errors relative to `gt_depth`: mean position error, trajectory RMSE, and final-frame error. For mapping with known trajectory, we report the number of frames and map points and perform qualitative inspection of the reconstructed polyhedron and supporting plane.

V. RESULTS AND DISCUSSION

A. Classical RGB-ToF Fusion

Fig. 2 shows the main classical fusion results on frame 223 of `dod_1500us_20MHz`. The baseline nearest-neighbor upsample preserves the original ToF support but leaves strong sparsity, jagged boundaries, and missing regions around the object. RGB-guided weighted median filtering improves local robustness without strongly changing geometry, which explains why it remained one of the strongest conservative baselines in the overlap-only benchmark.

Confidence-aware completion increases valid support and fills part of the missing region around the polyhedron, but this additional coverage is accompanied by a slight increase in overlap-region error. Confidence completion followed by WMF further smooths the reconstructed map and improves visual continuity, although it also starts to bias local geometry. The confidence+multi-frame variant stabilizes the result temporally and produces denser support, while

TABLE I
REPRESENTATIVE OVERLAP-ONLY RMSE ON `dod_1500us_20MHz`.
LOWER IS BETTER.

Method	Overlap RMSE
Baseline nearest ToF	107.51
WMF tradeoff	107.17
Confidence completion tradeoff	108.56
Confidence+WMF tradeoff	112.58
Multi-frame fusion tradeoff	112.10
Superpixel+plane tradeoff	106.38
Variational WLS (best reproduced)	103.72

the full confidence+multi-frame+superpixel-to-plane pipeline yields the visually densest classical reconstruction among the reproduced methods.

The variational WLS result in Fig. 2 is qualitatively different from the local filtering methods. Instead of only refining existing pixels, it imposes a stronger global smoothness prior guided by confidence and RGB structure. This produces a much denser and visually regular depth field, but its behavior in the benchmark and in SLAM must be interpreted separately from its visual appearance.

Representative overlap-only depth scores are summarized in Table I. Among the reproduced classical methods, the strongest conservative trade-off was achieved by superpixel-plane and WMF-based methods, while completion-heavy pipelines improved coverage more than overlap-only accuracy. Variational WLS achieved the best reproduced overlap-only score after parameter adaptation, confirming that global regularization can be very effective at the depth-reconstruction level.

B. Neural Depth Priors

Figs. 3 and 4 compare neural depth priors on two different scenes: the polyhedron scene and a staircase-like structured scene. These figures show that the neural methods behave very differently from the classical RGB-ToF fusion pipelines. Instead of refining noisy active depth, they infer dense depth mainly from learned monocular structure priors.

On the polyhedron scene in Fig. 3, DepthAnything produced the most geometrically plausible object among the tested neural methods, while AnyDepth failed to recover the correct object shape and instead produced a broad smooth depth gradient. SelfToF and DELTAR reconstructed denser structure than AnyDepth, but their object contours and local depth transitions remained less accurate than the ground truth. On the staircase scene in Fig. 4, DepthAnything again preserved the global scene layout better than the other neural baselines, whereas AnyDepth remained overly smooth and SelfToF / DELTAR showed limited structural fidelity.

Quantitatively, the neural methods were less competitive than the strongest classical and variational methods on the benchmarked `dod_1500us_20MHz` sequence. Their main weakness in this project is that visually plausible monocular depth does not necessarily agree with the metric geometry required for SLAM. This is especially important in our setup, where the downstream task is not only dense depth prediction but also geometric registration and mapping.

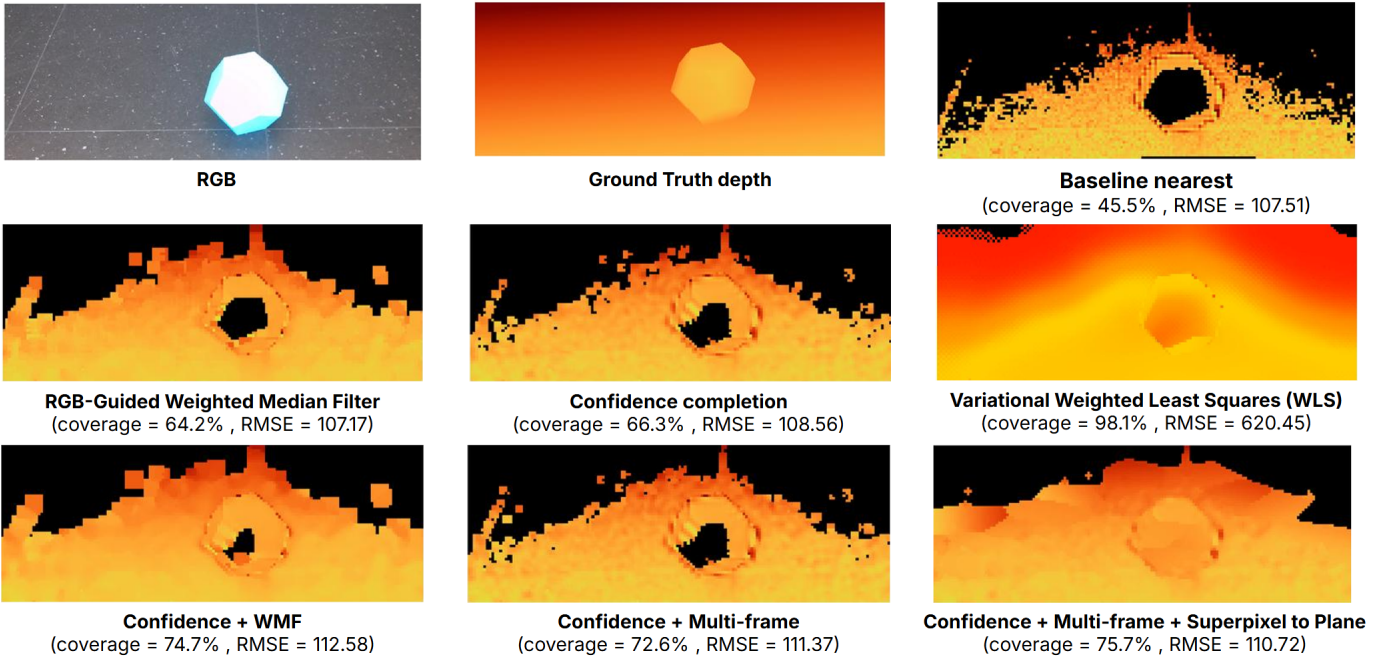


Fig. 2. Comparison of classical RGB-ToF fusion methods on frame 223 of `dod_1500us_20MHz`. The figure includes RGB, ground-truth depth, the nearest-neighbor baseline, RGB-guided WMF, confidence completion, variational WLS, confidence+WMF, confidence+multi-frame fusion, and confidence+multi-frame+superpixel-to-plane refinement.

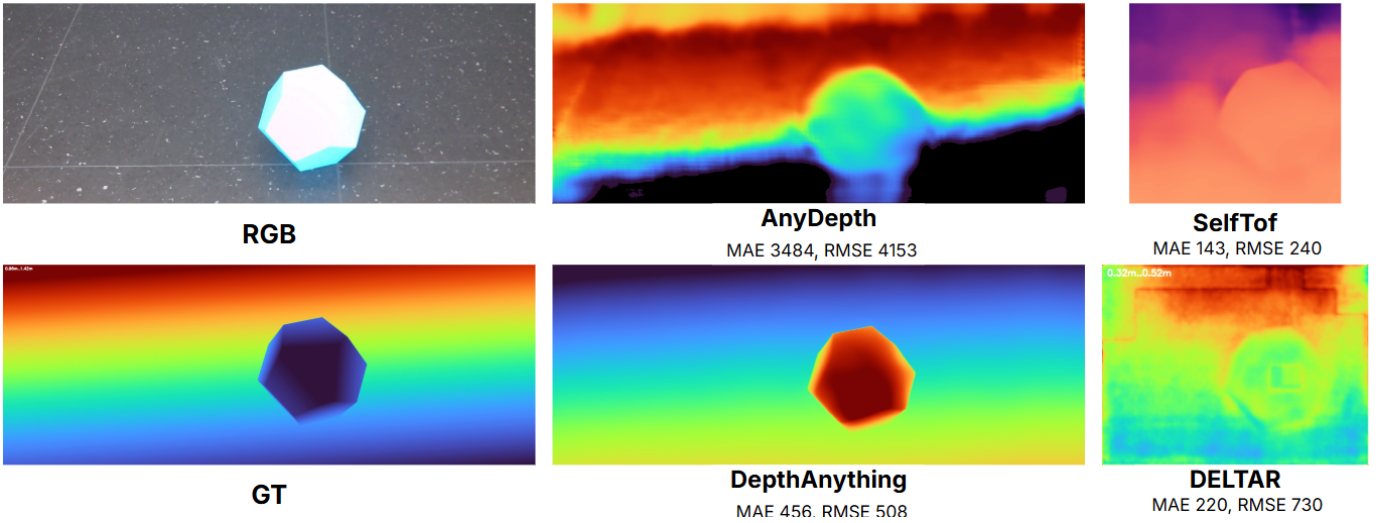


Fig. 3. Neural depth priors on the polyhedron scene. The figure compares AnyDepth, SelfToF, DepthAnything, and DELTAR against RGB and ground-truth depth.

C. SLAM Performance

The SLAM comparison is shown in Fig. 5, and the strongest trajectories are summarized in Table II. Here the ranking differs from the pure depth benchmark. The most accurate SLAM trajectories were achieved not by the visually densest methods, but by methods that preserved geometry conservatively enough for stable registration.

Fig. 5 makes this difference visually clear. The nearest-neighbor baseline remains sparse, but it preserves the coarse supporting plane reasonably well. Confidence com-

pletion+WMF produces a denser and more coherent reconstruction and substantially improves the trajectory relative to the raw baseline. Confidence+multi-frame+superpixel gives an even denser map, but with stronger geometric deformation in the accumulated structure. Variational WLS produces a smooth global map, yet this regularization does not automatically translate into stronger SLAM performance. Finally, the neural methods, especially AnyDepth, produce maps and trajectories that are much less consistent with the GT-based reference than the strongest classical geometric methods.

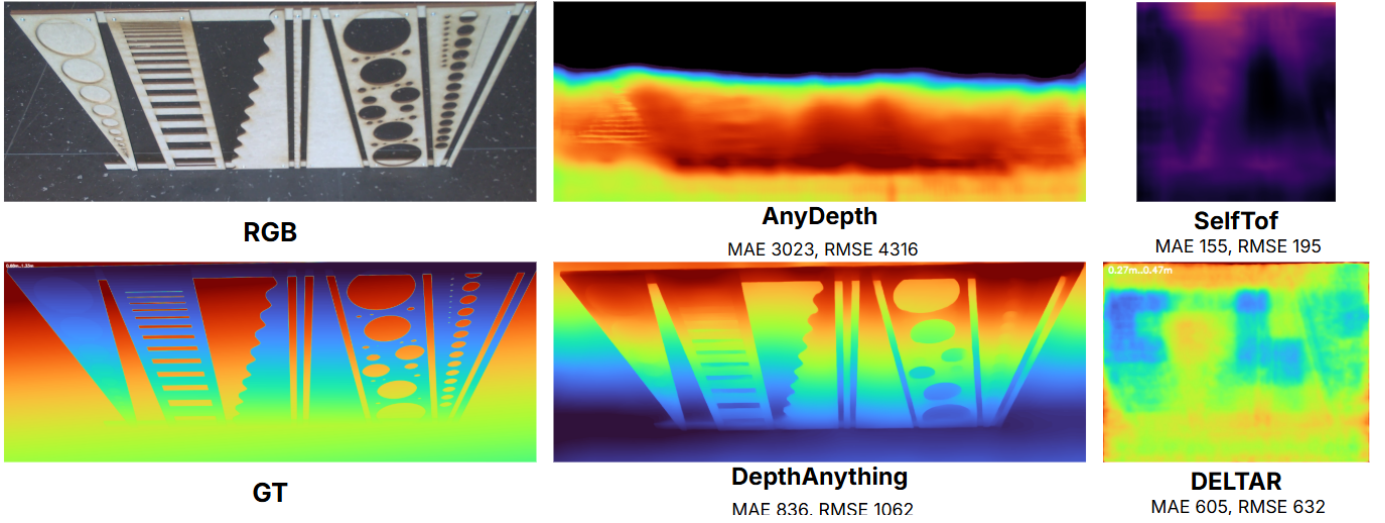


Fig. 4. Neural depth priors on the staircase scene. The figure compares AnyDepth, SelfToF, DepthAnything, and DELTAR against RGB and ground-truth depth.

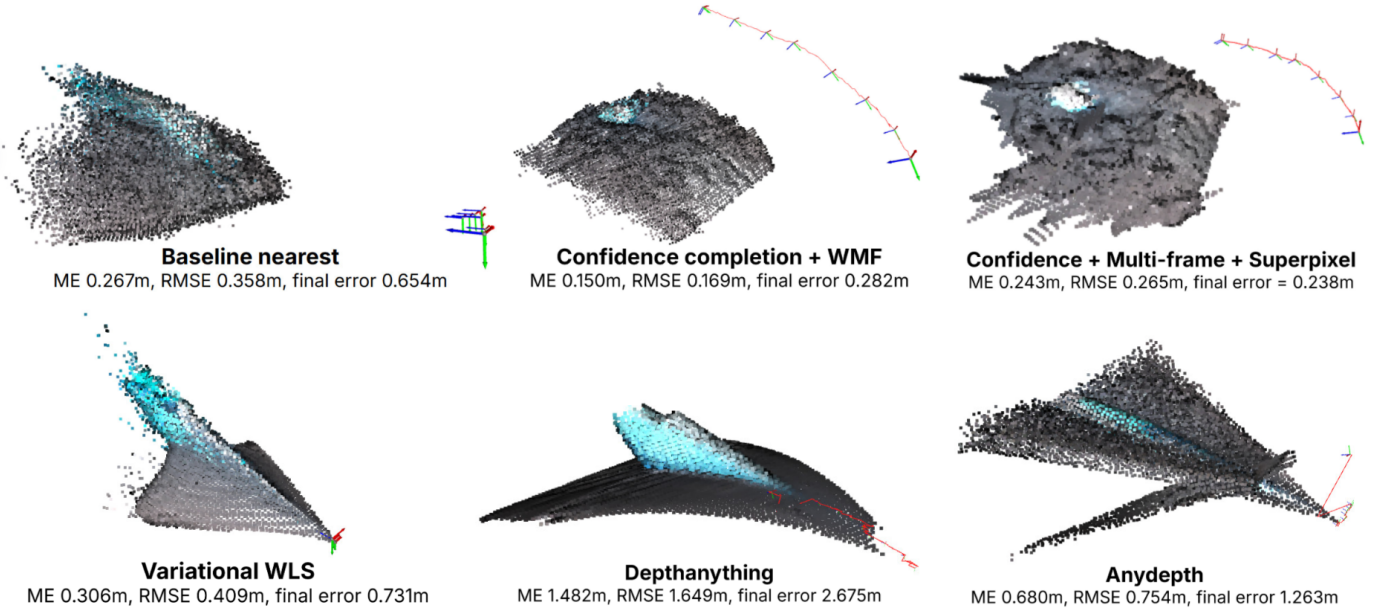


Fig. 5. Representative SLAM results. The figure compares reconstructed maps and trajectories for the nearest-neighbor baseline, confidence completion+WMF, confidence+multi-frame+superpixel, variational WLS, DepthAnything, and AnyDepth.

TABLE II
BEST SLAM TRAJECTORIES ON DOD_1500US_20MHZ, EVALUATED
AGAINST THE GT_DEPTH SLAM TRAJECTORY.

Method	Mean err.	RMSE	Final err.
tof_superpixel_plane	0.149 m	0.199 m	0.359 m
tof_confidence_plane	0.150 m	0.169 m	0.282 m
Confidence + Multiframe	0.243 m	0.265 m	0.238 m
Fusion + Superpixel to Plane			
Single ToF sensor	0.267 m	0.358 m	0.654 m

This is the central result of the project: the best depth reconstruction method is not automatically the best SLAM input.

Pixel-level density and visual completeness are not sufficient if the reconstructed geometry becomes biased enough to hurt frame-to-frame alignment.

D. Known-Trajectory Mapping

To isolate mapping quality from odometry drift, we additionally ran known-trajectory mapping, shown in Fig. 6. In this experiment, all methods use the same prescribed camera path, so differences in the reconstructed map are caused primarily by depth consistency rather than by pose-estimation errors.

The ground-truth result produces the sharpest supporting plane and the clearest polyhedron. The raw baseline already reconstructs the global scene layout, but the object is elongated

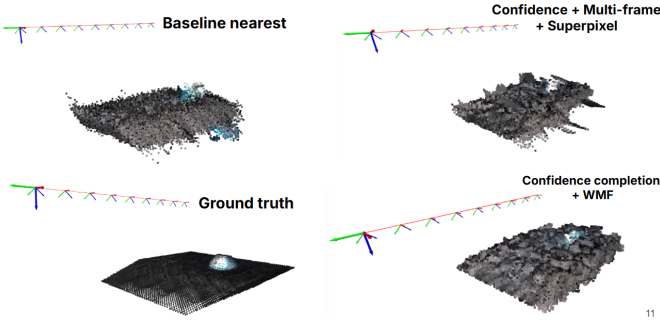


Fig. 6. Known-trajectory mapping results. Ground truth, raw baseline, and two fused depth variants are compared to isolate mapping quality from odometry drift.

and noisy. The confidence+multi-frame+superpixel native result increases density, but the object shape becomes more blurred and streaked. The confidence completion+WMF result produces a visually more regular volume, although it still remains less sharp than the GT reconstruction.

This experiment clarifies the difference between SLAM and mapping in the project. In SLAM, trajectory and map quality are coupled, so registration drift directly degrades the final reconstruction. In known-trajectory mapping, pose is fixed, so the remaining blur reflects depth inconsistency itself. This makes the mapping study a useful complement to the SLAM benchmark.

E. Discussion

Three main conclusions emerge from the experiments.

First, RGB-ToF fusion must be evaluated with more than one metric. Overlap-only RMSE is useful for measuring local refinement quality, but it does not capture whether a method preserves geometry well enough for registration and mapping.

Second, geometry-conservative methods were especially effective for SLAM. Superpixel-plane and confidence-plane variants did not always produce the densest maps, but they delivered the strongest trajectory accuracy because they preserved the planar support and object geometry more reliably.

Third, neural depth priors were visually interesting but less effective than the strongest classical and variational methods in this project. Their dense predictions did not consistently match the metric depth structure required by RGB-D odometry and point-cloud fusion.

Overall, the results show that RGB-ToF fusion must be treated not only as a depth-restoration problem, but as part of a full robotics perception pipeline.

VI. CONCLUSION

This project presented a comparative evaluation of RGB-ToF fusion methods with a specific focus on downstream robotics perception. We studied classical local filters, confidence-guided completion, temporal fusion, plane-based refinement, variational weighted least squares, and neural monocular-depth priors. These methods were benchmarked against ground-truth depth and then evaluated in an Open3D

RGB-D SLAM pipeline and in a known-trajectory mapping setting.

The main result is that the best depth-reconstruction method and the best SLAM method are not necessarily the same. Variational WLS achieved the strongest reproduced overlap-only depth benchmark, while geometry-conservative plane-based methods produced the most accurate SLAM trajectories. This confirms that RGB-ToF fusion must be assessed not only in terms of dense depth quality, but also in terms of geometric consistency for registration and mapping.

The experiments also showed that neural monocular-depth priors can generate visually plausible dense depth maps, but in the current project they remained less suitable for metric RGB-D SLAM than the strongest classical geometric methods. Overall, the study demonstrates that RGB-ToF fusion should be treated as a full perception-pipeline problem rather than only as an image restoration task.

Future work should combine motion-compensated temporal fusion, stronger uncertainty modeling, and a more advanced SLAM backend with loop closure and global trajectory optimization.